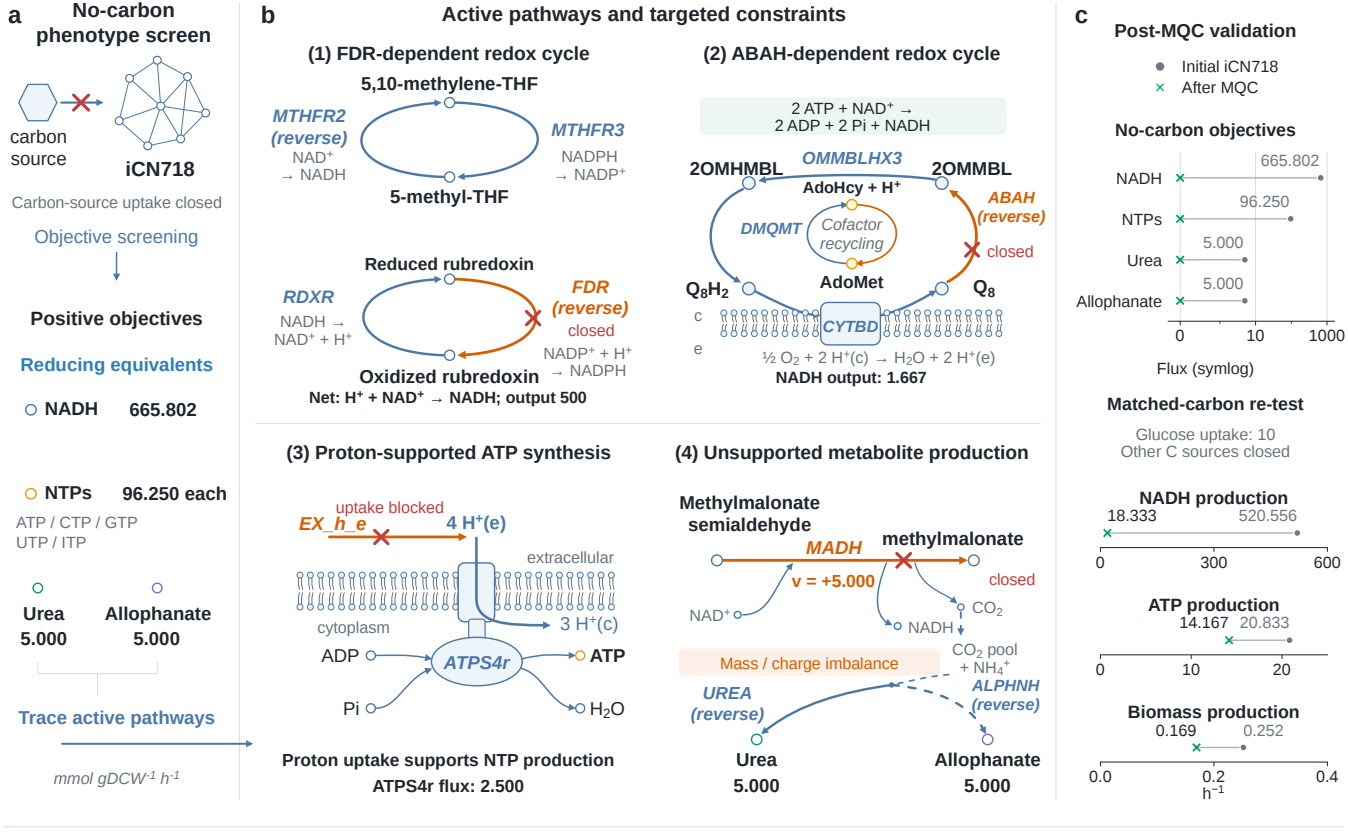

a
No-carbon phenotype screen
carbon source
iCN718
Carbon-source uptake closed
Objective screening
Positive objectives
Reducing equivalents
NADH 665.802
NTPs 96.250 each
ATP / CTP / GTP
UTP / ITP
Urea 5.000
Allophanate 5.000
Trace active pathways
mmol gDCW-1 h-1
b
Active pathways and targeted constraints
(1) FDR-dependent redox cycle
5,10-methylene-THF
MTHFR2 (reverse)
NAD+ → NADH
MTHFR3
NADPH → NADP+
5-methyl-THF
Reduced rubredoxin
RDXR
NADH → NAD+ + H+
FDR (reverse)
closed
NADP+ + H+ → NADPH
Oxidized rubredoxin
Net: H+ + NAD+ → NADH; output 500
(2) ABAH-dependent redox cycle
2 ATP + NAD+ → 2 ADP + 2 Pi + NADH
2OMHMBL
OMMBLHX3
2OMMBL
AdoHcy + H+
ABAH (reverse)
DMQMT
Cofactor recycling
closed
AdoMet
Q8H2
Q8
c
CYTBD
e
½ O2 + 2 H+(c) → H2O + 2 H+(e)
NADH output: 1.667
(3) Proton-supported ATP synthesis
uptake blocked
EX_h_e
4 H+(e)
extracellular
cytoplasm
3 H+(c)
ADP
ATPS4r
ATP
Pi
H2O
Proton uptake supports NTP production
ATPS4r flux: 2.500
(4) Unsupported metabolite production
Methylmalonate semialdehyde
MADH
methylmalonate
v = +5.000
closed
NAD+
CO2
NADH
Mass / charge imbalance
CO2 pool + NH4+
UREA (reverse)
ALPHNH (reverse)
Urea 5.000
Allophanate 5.000
c
Post-MQC validation
Initial iCN718
After MQC
No-carbon objectives
NADH 665.802
NTPs 96.250
Urea 5.000
Allophanate 5.000
0
10
1000
Flux (symlog)
Matched-carbon re-test
Glucose uptake: 10
Other C sources closed
NADH production
18.333
520.556
0
300
600
ATP production
14.167
20.833
0
10
20
Biomass production
0.169
0.252
0.0
0.2
0.4
h-1
Active pFBA route
Problematic reaction
MQC stop
Metabolite
Membrane
Model-derived link
Flux: mmol gDCW-1 h-1
c: cytoplasm; e: extracellular